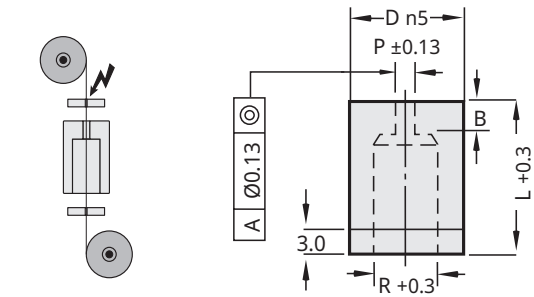


Code: **MKB**

Matrix with Counter-bore Relief

Blank type & with starting holes for wire-edm ISO 8977



With starting holes for "wir e-edm".

D	P	B - point length		R	L
		S (standard)	A (optional)		
8	0.8	4	-	4	20
10	0.8	4	8	6	
13	1.6	5	8	8	22
16	1.6	5	8	9.5	
20	1.6	5	12	12	25
22	1.6	6	12	15	
25	1.6	6	12	17.5	28
28	1.6	6	12	17.5	30
30	1.6	6	12	17.5	
32	1.6	6	12	21	32
35	1.6	6	12	21	35
38	1.6	8	12	27	40
40	1.6	8	12	27	
45	3.2	8	12	36	60
50	3.2	8	12	41	

Order: **MKB.** D x L - B

Order Example: **MKB. 20x25-S**

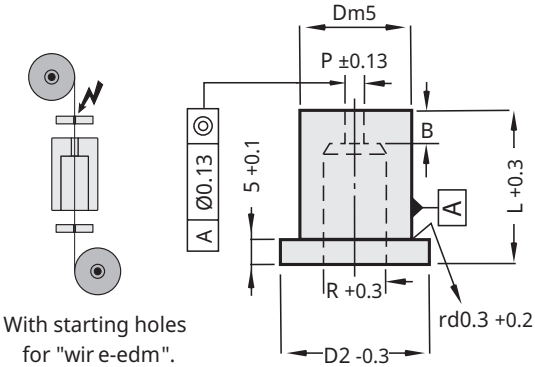
Material: 1.3343 (M2)
Hardness: 60 - 62 HRC



Code: **MLB**

Headed Matrix with Counter-bore Relief

Blank type & with starting holes for wire-edm ISO 8977



With starting holes for "wir e-edm".

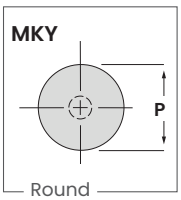
D	D2	P	B - point length		R	L
			S (standard)	A (optional)		
8	11	0.8	4	-	4	20
10	13	0.8	4	8	6	
13	16	1.6	5	8	8	22
16	19	1.6	5	8	9.5	
20	23	1.6	5	12	12	25
22	25	1.6	6	12	15	28
25	28	1.6	6	12	17.5	30
28	31	1.6	6	12	17.5	32
30	33	1.6	6	12	17.5	
32	35	1.6	6	12	21	35
35	38	1.6	6	12	21	40
38	41	1.6	8	12	27	
40	43	1.6	8	12	27	

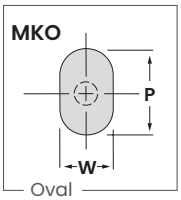
Order: **MLB.** D x L - B

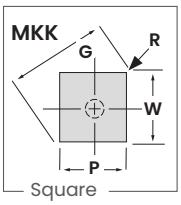
Order Example: **MLB. 16x20-S**

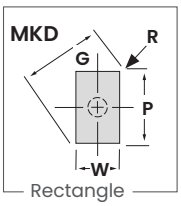
Material: 1.3343 (M2)
Hardness: 60 - 62 HRC

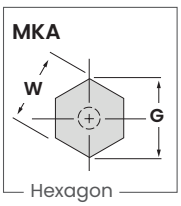
SHAPES

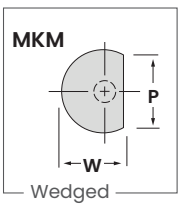


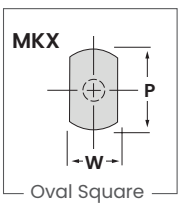


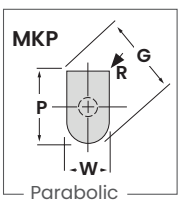


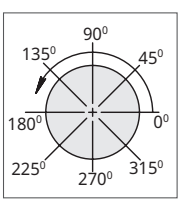




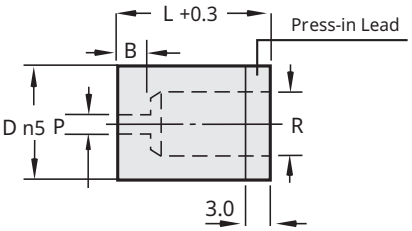












Code: **MK..**

Please refer to the table on the left side

MKY - MKO - MKK - MKD - MKA - MKM - MKX - MKP

Matrix - Straight ISO 8977

Apart from our standards, special dimensions and complex geometries can be produced according to your technical drawings.

Code	D	B - point length		ROUND (MKY)	SHAPE		R	L								
		S (standard)	A (optional)	Range P	Min. W	Max. P/G		20	22	25	28	30	32	35	40	60
MK...08	8	4	-	1.50 - 3.20	1.50 - 3.20		4	•	•	•	•	•	•	•		
MK...10	10	4	8	1.60 - 5.00	1.60 - 5.00		6	•	•	•	•	•	•	•		
MK...13	13	5	8	1.80 - 7.20	1.80 - 7.20		8	•	•	•	•	•	•	•		
MK...16	16	5	8	5.00 - 8.80	2.50 - 8.80		9.5	•	•	•	•	•	•	•		
MK...20	20	5	12	5.50 - 11.00	3.20 - 11.00		12	•	•	•	•	•	•	•	•	
MK...22	22	6	12	7.50 - 14.00	4.00 - 14.00		15	•	•	•	•	•	•	•	•	
MK...25	25	6	12	9.50 - 16.50	4.80 - 16.50		17.5	•	•	•	•	•	•	•	•	
MK...28	28	6	12	11.50 - 19.50	4.80 - 20.00		17.5	•	•	•	•	•	•	•	•	
MK...30	30	6	12	11.50 - 20.00	4.80 - 20.00		17.5	•	•	•	•	•	•	•	•	
MK...32	32	6	12	13.00 - 20.00	5.50 - 20.00		21	•	•	•	•	•	•	•	•	
MK...35	35	6	12	13.00 - 23.00	5.50 - 23.00		21	•	•	•	•	•	•	•	•	
MK...38	38	8	12	16.00 - 26.00	6.40 - 26.00		27	•	•	•	•	•	•	•	•	
MK...40	40	8	12	16.00 - 26.00	6.40 - 26.00		27			•	•	•	•	•	•	
MK...45	45	8	12	19.00 - 35.00	8.00 - 35.00		36						•		•	•
MK...50	50	8	12	22.00 - 40.00	9.00 - 40.00		41						•		•	•

Order:
MK (type) D x L x **shape** (P/W/R/G)

Material: 1.3343 (M2)
Hardness: 60 - 62 HRC

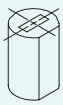
Order Example

MKO 25- (A) 32 - P=13.0 W=10.5 - K1

Code | *D* | *L* | *Shape* | *Keying Info*

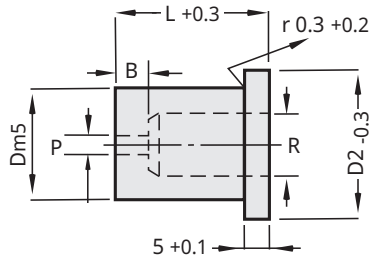
** "B" Point length*

* If you want the **Point Length** to be Optional, add code **(A).** No code is written in the standard.



For details on Keying Information / Key Flat Options, and More Complex Special Shapes, please refer to the beginning of Category 6.





Code: **ML..**

Matrix - Headed ISO 8977

Please refer to the table on the right side
MLY - MLO - MLK - MLD - MLA - MLM - MLX - MLP

Apart from our standards, special dimensions and complex geometries can be produced according to your technical drawings.

Code	D	D2	B - point length		ROUND (MLY)	SHAPE		R	L							
			S (standard)	A (optional)	Range P	Min. W	Max. P/G		20	22	25	28	30	32	35	40
ML...08	8	11	4	-	1.50 - 3.20	1.50 - 3.20		4	•	•	•	•	•	•	•	
ML...10	10	13	4	8	1.60 - 5.00	1.60 - 5.00		6	•	•	•	•	•	•	•	
ML...13	13	16	5	8	1.80 - 7.20	1.80 - 7.20		8	•	•	•	•	•	•	•	
ML...16	16	19	5	8	5.00 - 8.80	2.50 - 8.80		9.5	•	•	•	•	•	•	•	
ML...20	20	23	5	12	5.50 - 11.00	3.20 - 11.00		12	•	•	•	•	•	•	•	•
ML...22	22	25	6	12	7.50 - 14.00	4.00 - 14.00		15	•	•	•	•	•	•	•	•
ML...25	25	28	6	12	9.50 - 16.50	4.80 - 16.50		17.5	•	•	•	•	•	•	•	•
ML...28	28	31	6	12	11.50 - 19.50	4.80 - 20.00		17.5	•	•	•	•	•	•	•	•
ML...30	30	33	6	12	11.50 - 20.00	4.80 - 20.00		17.5	•	•	•	•	•	•	•	•
ML...32	32	35	6	12	13.00 - 20.00	5.50 - 20.00		21	•	•	•	•	•	•	•	•
ML...35	35	38	6	12	13.00 - 23.00	5.50 - 23.00		21	•	•	•	•	•	•	•	•
ML...38	38	41	8	12	16.00 - 26.00	6.40 - 26.00		27	•	•	•	•	•	•	•	•
ML...40	40	43	8	12	16.00 - 26.00	6.40 - 26.00		27			•	•	•	•	•	•

Order:
ML (type) D x L x **shape** (P/W/R/G)

Material: 1.3343 (M2)
Hardness: 60 - 62 HRC

Order Example

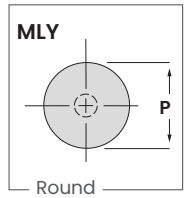
MLY 25- (A) 32 - P=13.0
Code d2 L1 Shape
** "B" Point length*

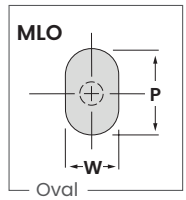
* If you want the **Point Length** to be Optional, add code **(A).** No code is written in the standard.

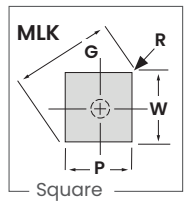


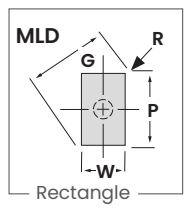
For details on Keying Information / Key Flat Options, and More Complex Special Shapes, please refer to the beginning of Category 6.

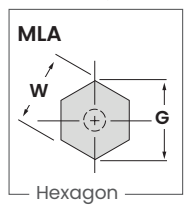
SHAPES

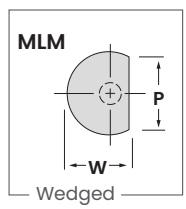


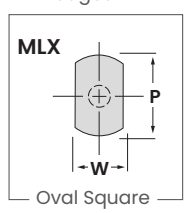


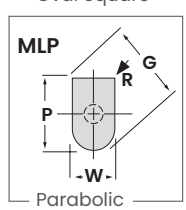


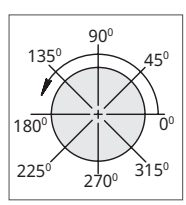




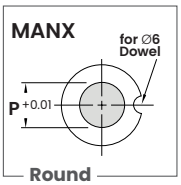


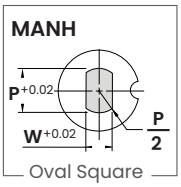


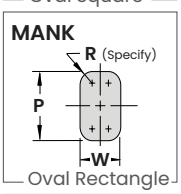


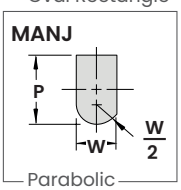


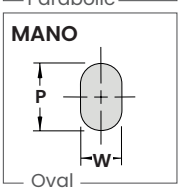
SHAPES

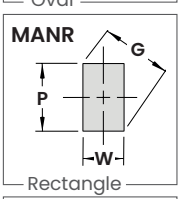


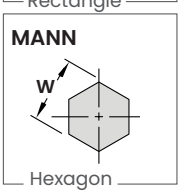


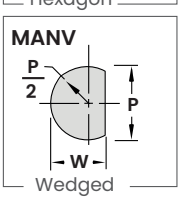


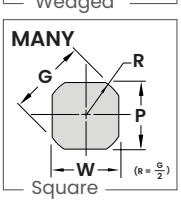


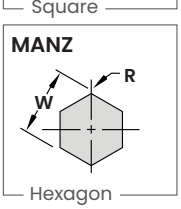




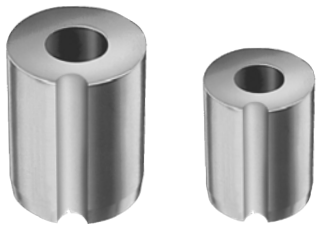


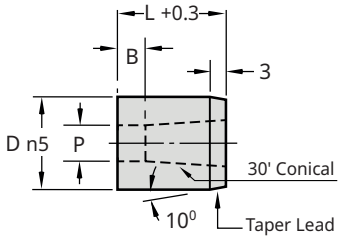






NEW





Code: **MAN..**

Matrix - Tapered Relief **Ref.: NAAMS**

This series of matrix comes standard with the **'PK6' Dowel Grooves** (NAAMS standard) option for Ø6 dowels, unless otherwise specified.

Please refer to the table on the left side

MANX - MANH - MANK - MANJ - MANO - MANR - MANN - MANV - MANY - MANZ

Code	D	B - point length		ROUND	SHAPE		L								
		S (standard)	A (optional)	Range P	Min. W	Max. P/G	13	16	20	22	25	28	30	32	35	40
MAN.. 10	10	4	5	1.60 - 6.80	1.30 - 6.80		•	•	•	•	•	•	•	•	•	
MAN.. 13	13	5	8	3.00 - 8.80	1.90 - 8.80		•	•	•	•	•	•	•	•	•	
MAN.. 16	16	5	8	7.40 - 10.80	1.90 - 10.80				•	•	•	•	•	•	•	
MAN.. 20	20	5	10	9.50 - 13.60	1.90 - 13.60				•	•	•	•	•	•	•	
MAN.. 22	22	6	10	10.50 - 15.00	1.90 - 15.00				•	•	•	•	•	•	•	
MAN.. 25	25	6	10	12.00 - 17.00	1.90 - 17.00				•	•	•	•	•	•	•	
MAN.. 32	32	6	12	16.00 - 22.00	1.90 - 22.00				•	•	•	•	•	•	•	
MAN.. 38	38	8	12	18.00 - 27.00	1.90 - 27.00				•	•	•	•	•	•	•	
MAN.. 40	40	8	12	18.00 - 27.00	1.90 - 27.00				•	•	•	•	•	•	•	
MAN.. 45	45	8	12	18.00 - 35.00	2.40 - 35.00					•	•	•	•	•	•	•
MAN.. 50	50	8	12	18.00 - 40.00	4.00 - 40.00					•	•	•	•	•	•	•

Order:
MAN (type) D x L x **shape** (P/W/G) x PK6

Material: 1.3343 (M2)
Hardness: 60 - 62 HRC

Order Example

MANX 25- (A) 32 - P=13.0 - PK6

Code | *D* | *L* | *Shape* | *Keying Info*

** "B" Point length*

* If you want the **Point Length** to be Optional, add code **(A).** No code is written in the standard.

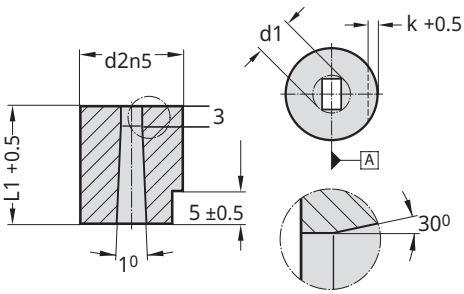
For details on Keying Information / Key Flat Options, and More Complex Special Shapes, please refer to the beginning of Category 6.

MAN matrixes conform to NAAMS™ standard for Tapered Relief Die Buttons.



Code: **MP..**

Matrix, Straight - Tapered Relief



Please refer to the table on the right side

MPY - MPO - MPK - MPD - MPA - MPM - MPX - MPP

d1 = shape	d2	k	L1
1.6 ~ 3.2	8	1	16
2 ~ 5	10		
3 ~ 7	13	1.5	19
5 ~ 8	16		22
7 ~ 11	20		25
11 ~ 16	25	2.5	28
16 ~ 19	32		32
19 ~ 28	40		

Order: **MP** (type) x d2 x L1
shape (P/W/R/G)

Material: 1.3343 (M2)
Hardness: 60 - 62 HRC

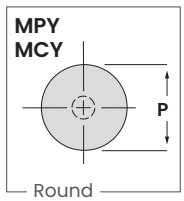
Order Example

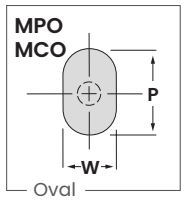
MPX 25-28 P=16.0 W=12.5 - K1/90°

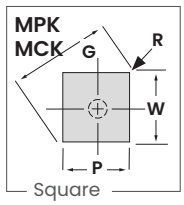
Code d2 L1 Shape Keying Info

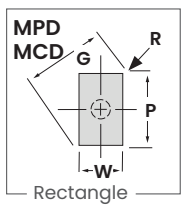
For details on Keying Information / Key Flat Options, please go to the beginning of Category-6.

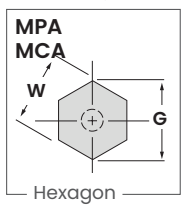
SHAPES

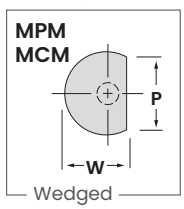


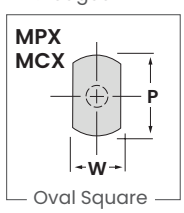


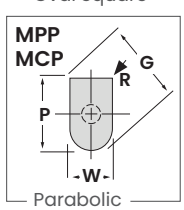


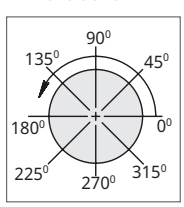








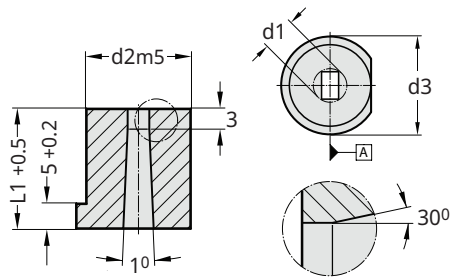






Code: **MC..**

Matrix, Headed - Tapered Relief



Please refer to the table on the left side

MCY - MCO - MCK - MCD - MCA - MCM - MCX - MCP

d1 = shape	d2	d3	k	L1
1.6 ~ 3.2	8	11	1	16
2 ~ 5	10	13		
3 ~ 7	13	16	1.5	19
5 ~ 8	16	19		22
7 ~ 11	20	23		25
11 ~ 16	25	28	2.5	28
16 ~ 19	32	35		32
19 ~ 28	40	43		

Order: **MC** (type) x d2 x L1
shape (P/W/R/G)

Material: 1.3343 (M2)
Hardness: 60 - 62 HRC

Order Example

MCY 16-25 P=8.0

Code d2 L1 Shape

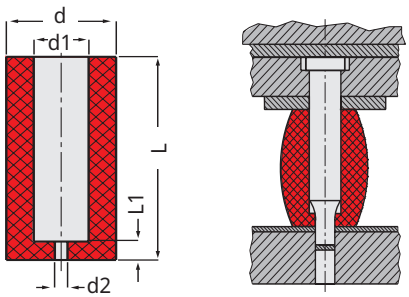
For details on Keying Information / Key Flat Options, please go to the beginning of Category-6.



Polyurethane Punch Stripper

Code: **PYB**

Overspread on punch (shock absorber)



Order Code	d	d1	d2	L1	L	Punch Length
PYB 445	4	17	1.6	5	45	56 / 63
PYB 655	6	19			55	63
PYB 855	8	21	3 Punch Hole			71
PYB 1055	10	23				80
PYB 1355	13	26				90
PYB 1655	16	30				
PYB 2055	20	38				100
PYB 2555	25	50				

d2: (1.6 - 3.0 mm) (d1) while opening hole diameter / drilling polyurethane (punch) bush should be applied and drilled in pressed (S max) position. Spring load is obtained while extending outwards.

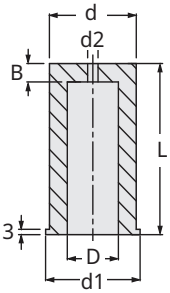
In S. max flexion, load coefficient data daN / Kg.

d1	17	19	21	23	26	30	38	50
S daN 3 mm	60	65	70	90	110	140	210	370
S daN 6 mm	115	120	130	160	190	230	360	650
S daN 9 mm	-	180	210	240	300	370	550	1020

S max: Load coefficient daN = Kg. / (10 newton), while polyurethane bush S max in pressed, load data is advisory.

In dies that polyurethane punch stripper is used, there is no need to dismantle stripper plate to make repair whetting and modification on die components, there is not any effect on precise parts, it is excellent for all painted / anodized, plastic plated and polished parts. It is compatible to use with oil and grease. It is overspread on punch. Placement is done according to the stripper hardness. There is no need for extra holder. Stepped-punch hole will be opened at the first stroke of press on stripper edge. **Especially, in large dies requiring very wide stripper plate, this product is very compatible.**





Heavy Duty Light Duty

Code: **MTS**

Polyurethane Stripper for Ball Lock

Exchangeable and compatible with NAAMS standards.

Punch Dia.	D	d	L	d1	d2	B
10	9.75	18	44	21	1.6	6
			54			
			64			
			74			
13	12.75	32	44	26	3	6
			54			
			64			
			74			
16	15.75	28	44	31	3	6
			54			
			64			
			74			
20	19.75	33	44	36	3	7
			54			
			64			
			74			
25	24.75	40	44	43	3	7
			54			
			64			
			74			
32	31.70	50	44	55	3	7
			54			
			64			
			74			
40	39.70	60	44	65	3	8
			64			

Order: **MTS.** d x L

Material: Polyurethane / 95 Shore-A

Polyurethane Length for Strippers:

While ordering, please specify punch length.

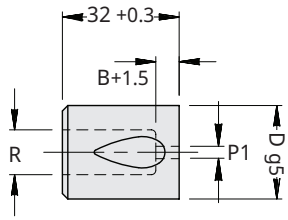
Heavy Duty Ball Lock Punch Length	Ligth Duty Ball Lock Punch Length	Recommended Stripper Length "L"
80	71	44
90	80	54
100	90	64
110	100	74

NEW



Ball Lock Matrix - Blank

Blank Type

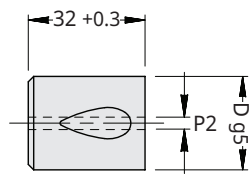


Material:
1.3343 (M2)

Hardness:
60 - 62 HRC

Code: **MBC**

Order Code	D	B	R	P1
MBC 13	13	4	5.8	1.2
MBC 16	16	5	8	1.6
MBC 20	20	5	11.9	2
MBC 25	25	6	16	3.6
MBC 32	32	6	20	4.4



Material:
1.3343 (M2)

Hardness:
60 - 62 HRC

Code: **MBE**

Order Code	D	P2
MBE 13	13	1.2
MBE 16	16	1.6
MBE 20	20	2
MBE 25	25	2.4
MBE 32	32	2.4

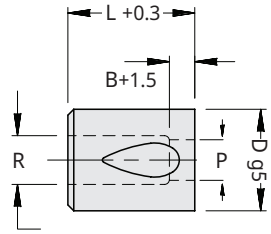
NEW



Ball Lock Matrix

Code: **MB..**

Automotive Standard / Shaped Types



Please refer to the table on the right side

MBY - MBO - MBK - MBD - MBA - MBM - MBX - MBP

Code	D	B	R	L	ROUND (MBY)	SHAPE	
					Range P	Min. W	Max. G/P
MB.. 13	13	4	6	32	1.6 - 5	1.6	5
MB.. 16	16	5	8	32	3.2 - 7.2	2	7.2
MB.. 20	20	5	12	32	4 - 11	2.5	11
MB.. 25	25	6	16	32	8 - 15	4	15
MB.. 32	32	6	20	32	11 - 19	5	19

Order:
MB. (type) D x L x Shape (P/W/G)

Material: 1.3343 (M2)
Hardness: 60 - 62 HRC

Order Example

MBY 25-32 P=13.0
Code D L Shape

For details on Keying Information / Key Flat Options, please go to the beginning of Category-6.

SHAPES

